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ADHD and Narcolepsy: Understanding the Connection

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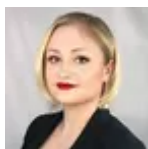
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Attention deficit hyperactivity disorder (ADHD) is a [neurodevelopmental disorder](#) characterized by hyperactivity, impulsivity, and inattention (difficulty paying attention). This condition may seem at odds with narcolepsy — a sleep disorder characterized by [excessive daytime sleepiness](#) and hypersomnia, among other [symptoms](#). However, the two conditions appear to be [related](#). In fact, studies have found a high prevalence of ADHD among people with narcolepsy — [as high as 30 percent](#).

Here, we will explore the connection between ADHD and narcolepsy, including similarities and differences in their symptoms and treatments. If you believe you may have ADHD with narcolepsy, talk to your health care provider. They will be able to help determine the cause of your symptoms and work with you to find the best course of treatment.

What Is ADHD?

ADHD is a chronic psychiatric disorder that affects an estimated [7.2 percent of adults](#) worldwide. ADHD may cause a person to have difficulty paying attention and controlling impulsive behaviors. A person with ADHD may also be overly active (hyperactive).

Signs of ADHD [include](#):

- Not listening when directly spoken to
- Being easily distracted and forgetful
- Fidgeting
- Interrupting or intruding on others

- Losing necessary items (like wallets, glasses, or keys)
- Talking excessively

ADHD often [begins in childhood](#) and persists into adulthood. However, in some people, the disorder is not diagnosed until adulthood. It may be challenging to recognize ADHD in adults, as the characteristic symptoms of the disorder — such as hyperactivity — may be less pronounced. ADHD symptoms may also [lessen as a person ages](#), or a person may learn to manage their symptoms with medication and coping strategies.

The Relationship Between ADHD and Narcolepsy

Research has indicated that [primary sleep disorders](#), including narcolepsy, can be [comorbid with ADHD](#). Comorbid conditions are conditions that can occur together in one person.

Adults with narcolepsy are [two times more likely](#) to have been diagnosed with ADHD in childhood than the general population. [One study](#) of children with narcolepsy found that as many as 15 percent of children with [type 1 narcolepsy \(with cataplexy\)](#) and 30 percent of children with type 2 narcolepsy (without cataplexy) had symptoms of ADHD. The symptoms of ADHD and narcolepsy can even overlap, potentially leading to misdiagnosis.

What Causes ADHD in Narcolepsy?

Researchers have suggested several potential causes of the comorbidity between ADHD and narcolepsy. Genetic analysis has revealed that iron metabolism, dopamine signaling, the immune system, and nervous system cells called glial cells are involved in [both narcolepsy and ADHD](#).

Dopamine and Noradrenaline Regulation

One possible explanation for the link between narcolepsy and ADHD is that the two [share common pathways](#) in the brain. Researchers have posited that ADHD results from problems regulating the neurotransmitter dopamine and the hormone noradrenaline. Noradrenaline dysregulation, in particular, can affect the rapid eye movement (REM) phase of sleep, similarly to the way [REM sleep is impacted](#) in people with narcolepsy. Medications used to treat ADHD also target the neurotransmitters used by the brain cells involved in REM sleep.

Low Iron Levels

Studies have found that low [iron levels](#) in the blood are associated with both ADHD and narcolepsy. Having lower iron levels has also been found to increase the severity of the symptoms of narcolepsy or ADHD.

Genetics

Researchers have suggested that there may be a genetic link between ADHD and narcolepsy. [Studies](#) have found that some people with ADHD and excessive daytime sleepiness have a shorter [REM latency](#) (time to reach the first phase of REM sleep after sleep onset) and even meet the [diagnostic criteria](#) for type 2 narcolepsy. These people often lack a specific genetic

trait seen in people with type 1 narcolepsy — and in some people with type 2 narcolepsy who don't have ADHD. This finding suggests that people with ADHD and excessive daytime sleepiness could be considered to have a unique form of type 2 narcolepsy.

How Are ADHD and Narcolepsy Treated?

While there is currently no cure for [narcolepsy](#) or [ADHD](#), a combination of medication and lifestyle changes can help manage their symptoms.

The following treatments may be helpful for both ADHD and narcolepsy. Talk to your health care team about the best way to manage your symptoms.

Stimulant Medications

Some medications are approved to treat narcolepsy but not ADHD, including [Xywav \(calcium, magnesium, potassium, and sodium oxybates\)](#), [Xyrem \(sodium oxybate\)](#), and [Wakix \(pitolisant\)](#). Other medications, called [stimulant drugs](#) (also known as psychostimulants), may be used to treat both ADHD and narcolepsy.

Psychostimulants are often the [first line of treatment](#) for ADHD. These medications help people with ADHD regulate their symptoms of hyperactivity and attention difficulties. Stimulant drugs have been prescribed to relieve excessive daytime sleepiness due to narcolepsy [since the 1930s](#).

These drugs, which include amphetamines and methylphenidates, help balance the levels of neurotransmitters in the brain. Common options for both ADHD and narcolepsy include:

- Adderall (amphetamine/dextroamphetamine)
- Dexedrine (dextroamphetamine)
- Ritalin or Concerta (methylphenidate)

[Provigil \(modafinil\)](#) and [Nuvigil \(armodafinil\)](#) are stimulant medications that help to improve wakefulness in people who have excessive daytime sleepiness associated with narcolepsy, obstructive sleep apnea, or shift work sleep disorder. Modafinil, in particular, has been found to [significantly improve](#) the symptoms of ADHD. It has also been used as an off-label medication to treat the disorder.

As one MyNarcolepsyTeam member shared, stimulants were very helpful in managing excessive daytime sleepiness: “I take Vyvanse in the mornings. It is also used to treat ADHD. If it wasn't for that, I would sleep my life away!”

Another member found benefits of taking more than one medication: “I was on Adderall, but it only helped me focus in the morning. I was still sleepy in the afternoon, even when taking it. I added Provigil, and after a week or so, it is great for me! Subtle, but no sleepiness!”

Talk to your doctor about your options — it may take some time for you and your health care team to find the best medication to treat your symptoms.

Good Sleep Hygiene

To treat narcolepsy, many neurologists recommend a combination of medications and lifestyle changes that support good sleep hygiene. As ADHD can also lead to [sleep disturbances](#), forming positive habits around sleeping may also be beneficial to people with ADHD.

The National Institute of Neurological Disorders and Stroke [recommends](#) that people with narcolepsy practice the following habits as much as possible:

- Stick to a regular sleep schedule, even on weekends.
- Take short naps when you feel sleepiest during the day.
- Exercise daily for at least 20 minutes.
- Avoid caffeine, alcohol, or heavy meals before bedtime.
- Don't smoke, especially in the evening.
- Keep your sleeping space comfortable and maintain a cool temperature.
- Take time to relax before bed.

Cognitive Behavioral Therapy

In addition to medications, psychotherapy has been shown to be effective in helping to treat narcolepsy and ADHD. A specific type of psychotherapy known as [cognitive behavioral therapy \(CBT\)](#) has been found to help people with narcolepsy better manage sleep patterns, manage anxiety, and control stimuli that trigger [cataplexy](#). CBT programs designed specifically for [people with ADHD](#) can help them more effectively manage their time, cope with difficulties in executive functioning, and regulate their emotions and impulsivity.

People with [ADHD](#) and [narcolepsy](#) are also at a higher risk of experiencing depression or anxiety. CBT can help these people manage anxiety or depressive disorders and their symptoms.

Meet Your Team

[MyNarcolepsyTeam](#) is the social network for people living with narcolepsy. Here, more than 8,000 members come together to ask questions, offer support and advice, and share stories with others who understand life with narcolepsy.

Have you been diagnosed with narcolepsy and ADHD? Share your story in the comments below or by posting on [MyNarcolepsyTeam](#).

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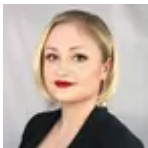




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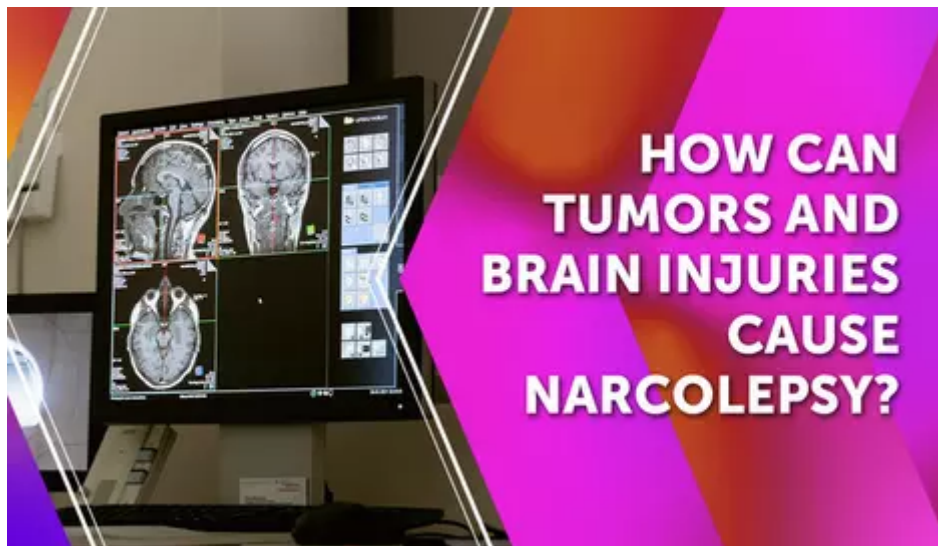


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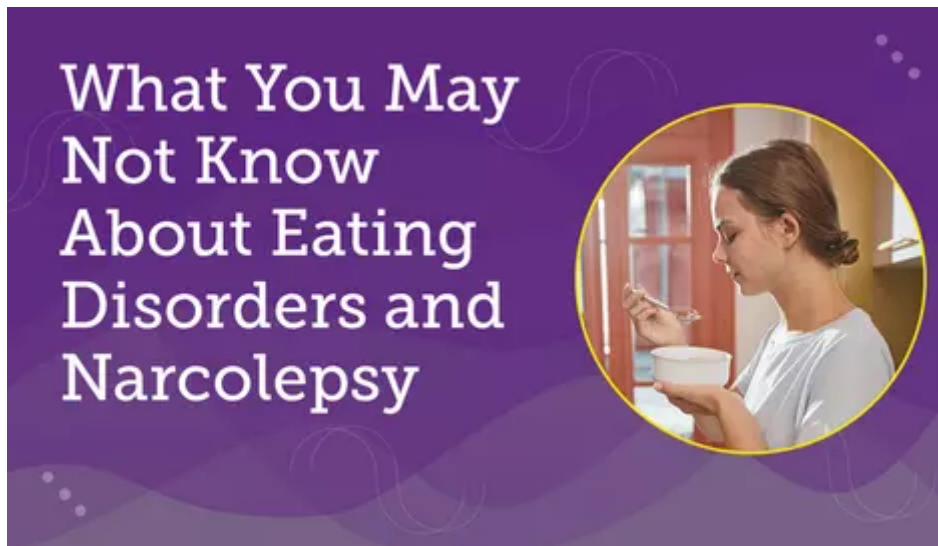
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